

## **Exercise 6: Simulation of a Cloud Scenario using CloudSim and Implementation of a Priority Scheduling Algorithm**

### **Aim**

To simulate a simple cloud scenario using CloudSim and implement a priority-based scheduling algorithm for executing cloud tasks.

### **Prerequisites**

- Java JDK
- Eclipse IDE
- CloudSim library

### **Procedure**

#### **Step 1: Create the Java Project**

Open Eclipse and create a Java project named "cloudsimscheduling". Add the CloudSim library to the project.

#### **Step 2: Create the Program**

Create a Java class named "Experiment6" inside the src folder.

#### **Step 3: Create Cloud Tasks**

Create four cloud tasks (Cloudlet 1, 2, 3 and 4) and assign priorities 2, 5, 1 and 4 respectively.

#### **Step 4: Apply Priority Scheduling**

Compare the priorities of the tasks and arrange them from highest to lowest priority. The task with the highest priority is scheduled first.

#### **Step 5: Execute the Simulation**

Run the program in Eclipse and display the scheduling order in the Console.

### **Program**

```
public class Experiment6 {
    public static void main(String[] args) {

        int[] task = {1, 2, 3, 4};
        int[] priority = {2, 5, 1, 4};

        for (int i = 0; i < 4; i++) {
            for (int j = i + 1; j < 4; j++) {

                if (priority[i] < priority[j])
                { int temp = priority[i];
                  priority[i] = priority[j];
                  priority[j] = temp;
                }
            }
        }
    }
}
```

```

        temp = task[i];
        task[i] = task[j];
        task[j] = temp;
    }
}

System.out.println("Priority Scheduling:");

for (int i = 0; i < 4; i++) {
    System.out.println( "Ta
        sk " + task[i] +
        " Priority = " + priority[i]
    );
}
}
}

```

## Output

Priority Scheduling:  
 Task 2 Priority = 5  
 Task 4 Priority = 4  
 Task 1 Priority = 2  
 Task 3 Priority = 1

*Figure: Priority scheduling program executed successfully in Eclipse*

## Result

Thus, the cloud scheduling scenario was successfully simulated and the priority-based scheduling algorithm was implemented. The tasks were arranged and scheduled according to their priority.

### Execution Screenshot:

